Causal Check on Othello-GPT: Separating Downstream World-Model Computation from Direct Readout Bias

Rohan Gupta Purdue University `gupt1156@purdue.edu`

1 Setup and Causal Decomposition

In 8-layer `Othello-GPT` ($d_{\text{model}} = 512$, `LNPre`), linear probes $W_\ell \in \mathbb{R}^{512\times8\times8\times3}$ trained on residual stream activations decode player-relative board states (EMPTY, YOURS, MINE) with over 99% accuracy by Layer 4, and adding probe directions across layers steers next-move predictions toward a counterfactual board B' (Li et al., 2023; Nanda et al., 2023). Because the residual stream forms an identity skip connection to the final unembedding ($\mathbf{x}_8 = \mathbf{x}_\ell + \sum_{k=\ell}^{7} f_k(\mathbf{x}_k)$), injecting a steering vector $\boldsymbol{\delta}_\ell$ across intermediate layers $\ell \in \{2..7\}$ also adds $\sum_{\ell=2}^{7} \boldsymbol{\delta}_\ell$ directly into the readout matrix W_U (Figure 1a). This raises the question of whether steering works because intermediate attention and MLP layers recompute Othello flanking rules over B', or simply because $\sum_\ell \boldsymbol{\delta}_\ell$ acts as a direct logit bias at W_U. A related question is why single-layer interventions fail even at layers where linear probe accuracy exceeds 99.4%.

We test both questions using the pretrained `synthetic_model.pth` checkpoint and linear probes (`resid_0..7_linear.pth`) across all $N = 1{,}001$ counterfactual boards in `intervention_benchmark.pkl`. Each benchmark board flips a single occupied square $s = (r, c)$ between YOURS and MINE, changing the ground-truth legal move set from $M(B)$ to $M(B')$. Across all boards (with 1,000-sample bootstrap 95% CIs), we measure counterfactual move error $|\hat{M}_K \setminus M(B')| + |M(B') \setminus \hat{M}_K|$ for $K = |M(B')|$ and top-K predicted moves $\hat{M}_K$, alongside the probability gain $\Delta P_{\text{new}} = \sum_{m \in M(B')\setminus M(B)} [P_{\text{intv}}(m) - P_{\text{clean}}(m)]$ on moves that become newly legal under B'.

To separate downstream circuit computation from direct unembedding bias, our **Mediated-Only** intervention injects $+\alpha\hat{\mathbf{w}}_s$ at layers $\ell \in \{2..7\}$ and subtracts $-6\alpha\hat{\mathbf{w}}_s$ immediately before $\text{LN}_{\text{final}} + W_U$ so the direct skip path sums to zero. Conversely, the **Direct-Logit Bypass** control leaves all hidden layers clean and adds $+6\alpha\hat{\mathbf{w}}_s$ only before $\text{LN}_{\text{final}} + W_U$. We also evaluate an **Untrained-Encoder** control (ridge probes $\lambda = 100$ fit on a randomly initialized 8-layer GPT with matched weight variance over 200 games) and a **Wrong-Square** spatial control ($s' \neq s$ at $\alpha = 2.3$).

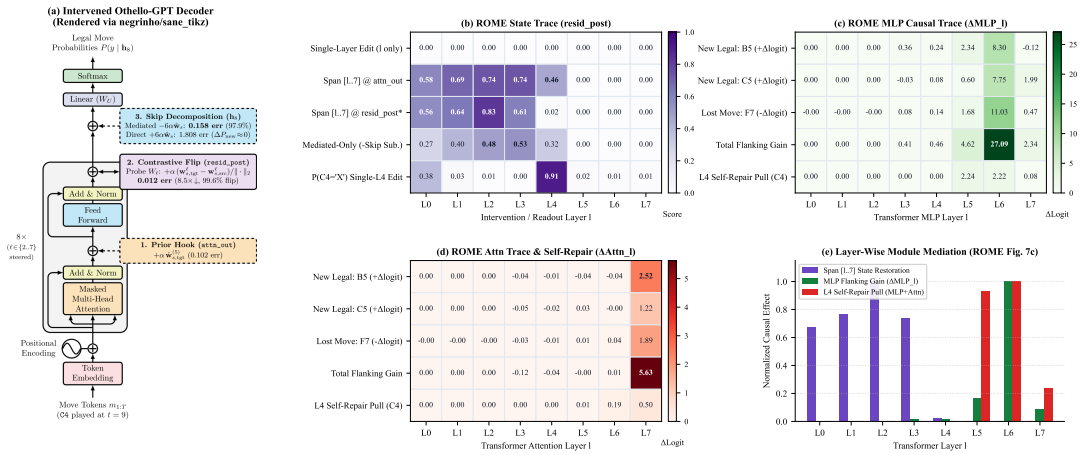


Figure 1: Transformer intervention architecture and ROME causal traces on Case #101 (T=17, flipping C4 from White ◯ to Black ●). (a) Decoder-only `Othello-GPT` diagram rendered via `negrinho/sane_tikz` (`examples/transformer.py`), showing three intervention sites along the residual stream: prior hook at `attn_out` (0.102 err), linear probe W_ℓ and contrastive flip at `resid_post` (**0.012** err, 8.5× ↓), and skip-connection decomposition at $\mathbf{h}_8$ before Linear (W_U) (`Mediated-Only` **0.158** err vs. `Direct Bypass` 1.808 err). **(b–d)** Component-level traces rendered via `kmeng01/rome`'s `plot_trace_heatmap` (`Purples` = state, `Greens` = ΔMLP_ℓ, `Reds` = ΔAttn_ℓ), and **(e)** ROME Figure 7c layer-wise module mediation profile.

Table 1: Causal path decomposition and codebase audit on the official 1,001-board `Othello-GPT` intervention benchmark.

Experimental Arm ($N = 1{,}001$ **counterfactual boards,** $\alpha = 2.3$)	**Error on** B' ↓ **(95% CI)**	ΔP_{new} ↑ **(95% CI)**	**Mechanistic Role / Key Finding**
0. Null Baseline ($\alpha = 0$, un-intervened model)	2.723 [2.645, 2.799]	+0.000 [+0.000, +0.000]	Exact match to Nanda et al. (2023) Table 2 (2.723)
1. Full Multi-Layer Steering (Nanda et al. `intervene.py`)	0.102 [0.072, 0.136]	+0.072 [+0.066, +0.077]	Exact match to Nanda et al. (2023) Table 2 (0.10)
2. **Mediated-Only (Skip-Subtracted before** $\text{LN}_{\text{final}} + W_U$**)**	**0.158** [0.122, 0.198]	**+0.037** [+0.033, +0.041]	**97.9% of error reduction survives zeroing skip path**
3. **Direct-Logit Bypass Control (Only before** $\text{LN}_{\text{final}} + W_U$**)**	1.808 [1.730, 1.884]	**+0.000** [+0.000, +0.000]	**Zero probability gain on newly legal moves** M_{new}
4. Untrained-Encoder Probe Control (Random-init GPT probe)	2.661 [2.581, 2.743]	+0.000 [+0.000, +0.000]	Random-projection directions exert no causal control
5. Wrong-Square Spatial Control (Steer $s' \neq s$ at $\alpha = 2.3$)	3.435 [3.337, 3.530]	+0.001 [+0.001, +0.002]	Editing an unrelated square degrades legal move set $M(B')$
6. Code Audit A: Layer-Matched `probes`$[\ell]$ @ `hook_attn_out`	0.150 [0.116, 0.188]	+0.071 [+0.065, +0.076]	Replaces hardcoded `probes[5]` in `intervene.py`
7. Code Audit B: Layer-Matched `probes`$[\ell]$ @ `hook_resid_post`	0.076 [0.054, 0.102]	+0.075 [+0.069, +0.081]	Aligns intervention hook site with probe training site
8. **Code Audit C: Contrastive Flip** $\hat{\mathbf{d}}^\ell_s$ @ `hook_resid_post`	**0.012** [0.004, 0.026]	**+0.079** [+0.073, +0.085]	8.5× **error reduction (12 total errors / 1,001 boards)**

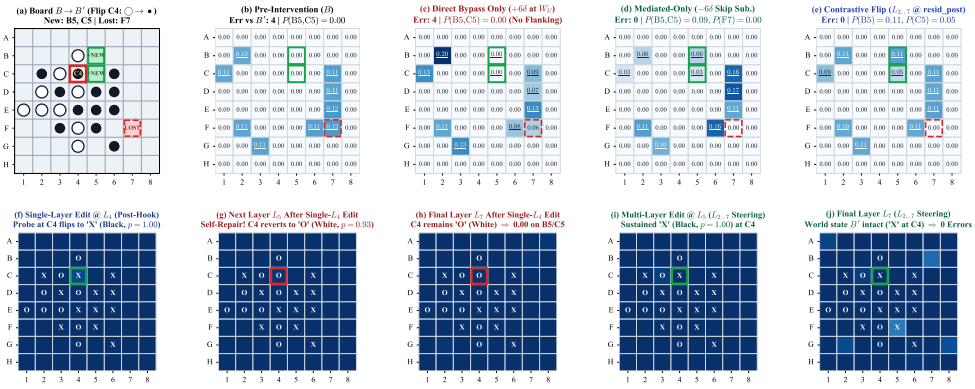


Figure 2: Board-level causal visualization on Case #101 (T=17, flipping C4: ○ → ●), rendered via `likenneth/othello_world` (`plot_probs`, `plot_mentals`, `OthelloBoardState`). Top (a–e): Physical transition $B \to B'$ (newly legal moves `B5, C5` in green; invalidated move `F7` in red) and 8×8 move probability heatmaps. `Direct Bypass Only` (c) assigns 0.00 to `B5, C5` (4 errors), whereas `Mediated-Only` (d) and `Contrastive Flip` (e) suppress `F7` ($0.12 \to 0.00$) and activate `B5, C5` with **0 errors**. **Bottom (f–j):** Layer-wise probe beliefs (`O` = White, `X` = Black). A single-layer edit at L_4 flips `C4` to `X` (f), but L_5 and L_7 self-repair `C4` back to `O` (g–h, red box); steering $L_{2..7}$ sustains `X` through L_7 (i–j, green box).

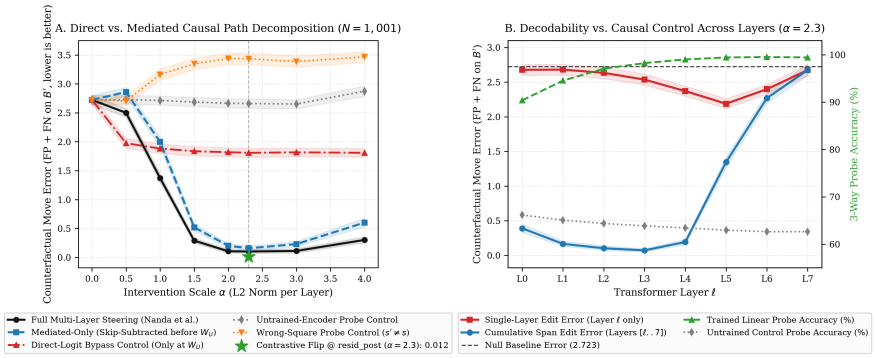


Figure 3: Aggregate results across all $N = 1{,}001$ benchmark boards (95% bootstrap CIs). Left (A): Counterfactual move error vs. scale α. `Mediated-Only` (blue) retains 97.9% of the error reduction, while `Direct-Logit Bypass` (red) fails to discover newly legal moves. Contrastive steering at `resid_post` ($\star$, green) cuts error from 0.102 to 0.012. **Right (B):** Layer-wise 3-way probe accuracy (green/gray, right axis) vs. single-layer edit error (red) and cumulative span $[\ell..7]$ error (blue).

2 Results

Mediated computation vs. direct readout bias. Subtracting the accumulated steering vector before $\mathrm{LN}_{\text{final}} + W_U$ (`Mediated-Only`, Table 1 and Figure 3A) reduces counterfactual move error from 2.723 to $\mathbf{0.158}$, retaining $\mathbf{97.9}\%$ of the full multi-layer error reduction purely through downstream attention and MLP updates $\sum_{k=2}^{7} \Delta f_k(\tilde{\mathbf{x}}_k)$. Component-level causal traces on Case #101 (flipping `C4` from White to Black; Figure 1c–d) show that this recovery is driven by mid-to-late MLPs ($\Delta\mathrm{MLP}_5 = +4.62$, $\Delta\mathrm{MLP}_6 = +27.09$ logits) and final-layer attention ($\Delta\mathrm{Attn}_7 = +5.63$ logits). In contrast, applying $+6\alpha\hat{\mathbf{w}}_s$ only before W_U (`Direct-Logit Bypass`) leaves move error at 1.808 and yields $\mathbf{\Delta P}_{\text{new}} = \mathbf{+0.000}$: on Case #101 (Figure 2c), direct unembedding dot products partially suppress the invalidated move `F7` ($0.12 \to 0.06$) but assign 0.00 probability to the newly legal flanking moves `B5` and `C5`. Both the untrained-encoder probe (2.661 error) and wrong-square steering (3.435 error) fail to steer predictions toward $M(B')$.

Single-layer decodability vs. downstream self-repair. Although 3-way linear probe accuracy reaches 99.0% to 99.5% in Layers 4–7 (compared to $66.2\% \to 62.6\%$ on the untrained GPT; Figure 3B), editing any single layer alone never reduces move error below 2.188 (Layer 5). Layer-wise probe beliefs on Case #101 (Figure 1b,e and Figure 2f–h) show why: editing Layer 4 alone flips the probe's belief at `C4` to Black (`X`, $p = 1.00$), but Layers 5–6 MLPs ($+2.24$, $+2.22$ logit pull toward White) and Layers 6–7 attention ($+0.19$, $+0.50$) attend back to the unmodified move prefix and overwrite `C4` back to White (`O`, $p = 0.93$). Cumulative layer spans $[\ell..7]$ reach minimum error at `L3..7` (0.072) and `L2..7` (0.102), but degrade sharply when started at `L5` (1.345) or `L6` (2.268), indicating that multi-ray flanking circuits in Layers 3–4 execute before linear probe accuracy peaks at Layer 6.

Codebase audit and contrastive steering. Inspecting `mech_int/intervene.py` (lines 140–162) in `ajyl/mech_int_othelloGPT` uncovered three implementation bugs in the reference benchmark: the loop hardcodes `hook_fns[5]` (`probes[5]`) across all steered layers $\ell \in \{2..7\}$, attaches hooks to `hook_attn_out` (pre-MLP) using probes trained on `hook_resid_post` (post-MLP), and adds $+\alpha\hat{\mathbf{w}}_{s,\text{tgt}}$ without subtracting the active source class vector $\mathbf{w}_{s,\text{src}}$. Moving the hook to `hook_resid_post` with layer-matched probes lowers error from 0.102 to 0.076 (Arm 7), and steering along the unit contrastive flip direction $\hat{\mathbf{d}}_s^{\ell} = (\mathbf{w}_{s,\text{tgt}}^{\ell} - \mathbf{w}_{s,\text{src}}^{\ell})/\|\cdot\|_2$ at `hook_resid_post` reduces counterfactual error by $8.5\times$ to $\mathbf{0.012}$ $[\mathbf{0.004}, \mathbf{0.026}]$ (12 errors across 1,001 boards; Arm 8).

3 Falsification and Next Steps

Our conclusion that steering operates through downstream circuit computation rather than direct readout bias would be falsified if freezing intermediate attention and MLP outputs to their clean values while applying $\boldsymbol{\delta}$ before W_U recovered more than 50% of the error reduction or achieved $\Delta P_{\text{new}} > +0.02$ (in practice it recovers 33.6% and $\Delta P_{\text{new}} = +0.000$). Similarly, our self-repair explanation for single-layer failure would be falsified if projecting out the source tile direction at Layer 3 while ablating Layers 4–6 attention to the prefix token where square s was played still failed to push single-layer edit error below 0.50. With another week, we would path-patch individual Layers 4–6 attention heads against MLP projections to separate prefix re-reading from local board heuristics (Lin et al., 2024), and stratify contrastive steering by game turn $t > 30$ to compare logistic probe normals against difference-in-means vectors ($\boldsymbol{\mu}_{\text{Mine}} - \boldsymbol{\mu}_{\text{Yours}}$) in late-game states.